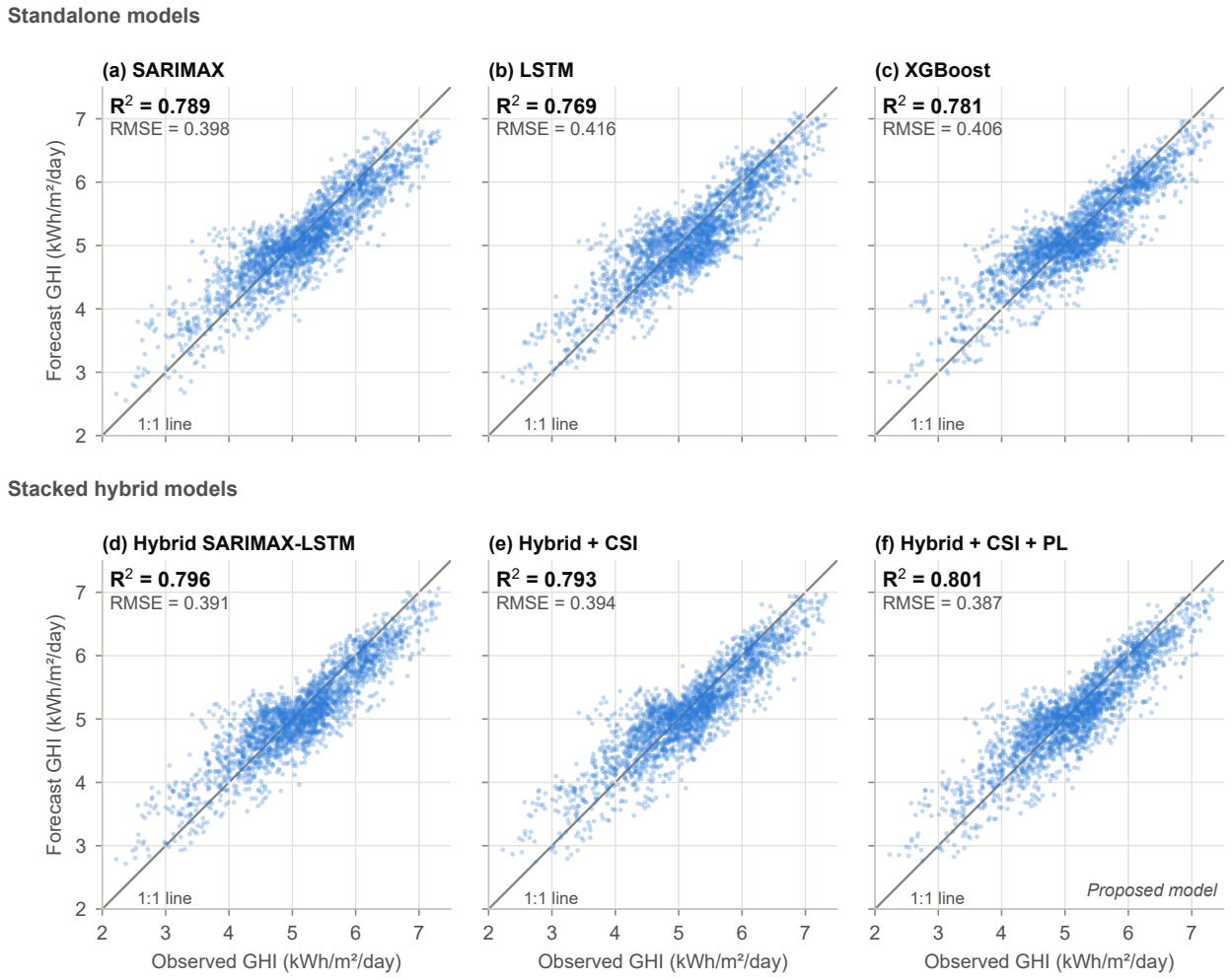

Standalone models
(a) SARIMAX
R² = 0.789
RMSE = 0.398
(b) LSTM
R² = 0.769
RMSE = 0.416
(c) XGBoost
R² = 0.781
RMSE = 0.406
Stacked hybrid models
(d) Hybrid SARIMAX-LSTM
R² = 0.796
RMSE = 0.391
(e) Hybrid + CSI
R² = 0.793
RMSE = 0.394
(f) Hybrid + CSI + PL
R² = 0.801
RMSE = 0.387
1:1 line
Proposed model
Forecast GHI (kWh/m²/day)
Observed GHI (kWh/m²/day)
2
3
4
5
6
7